

Development of Audiobook Interaction Design for People with Vision Impairment Based on User-Centered Design Approach

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Abstract—People with vision impairment have fewer sources of information than normal people because of the limitation to accessing information with their sight. As a result, this leads to an information gap. The use of audiobooks has become a way to lessen the gap. An audiobook is a recording of a book that is read aloud by the narrator. With the audiobook, a person can access the information contained in a book using his hearing. However, audiobooks have not been widely used and are still not common in developing countries, including Indonesia. This study proposes the development of an interaction design for an audiobook on the Android platform. The design is proposed using a User-Centered Design (UCD) approach that involves the predefined user in the development process. The design process begins with collecting data through questionnaires and interviews. From the data, we design two prototypes, i.e., a low-fidelity prototype and a high-fidelity prototype. These prototypes are tested on the predefined user to analyze the task completion rate, single ease question (SEQ), and system usability scale (SUS). The post-test questionnaire is additionally given to ensure the prototype has met the targeted usability and user experience goals, i.e., effective to use, easy to use, and helpful. Based on the evaluation of the test results, the proposed design has achieved the targeted usability and user experience goals.

Keywords—Audiobooks, blindness, interaction design, user-centered design, vision impairment.

I. INTRODUCTION

In 2020, around 1.1 billion people around the world have vision loss. Of those people, around 43.3 million people around 295 million were having moderate to severe vision impairment [1]. Because of such conditions, the sufferers face limitations in daily life, including the ability to obtain information.

People who are blind or have a severe visual impairment have a problem accessing information through visual means. The only way they can easily access information is through their hearing. Consequently, there will be an information gap between those who are visually impaired and those who are not.

There are a few ways to reduce the information gap. One of them is by using audiobooks, which is a recordings of a book

being read aloud in the form of a CD or internet-accessible [2]. By using audiobooks, visually impaired people can access information from a book through their hearing. Unfortunately, the use of audiobooks is still not common in Indonesia [3].

One way to increase the distribution of audiobooks in Indonesia is through mobile applications that allow access and usage of audiobooks. This is based on the number of smartphone users in Indonesia which has reached 183.68 million [4]. In addition, to make it more accessible for blind and visually impaired people, the application needs a few adjustments such as a label for elements, descriptive language [5], good contrast, and dark mode [6].

II. LITERATURE STUDY

A. User-Centered Design (UCD)

User-centered design is one of the approaches to designing a product in interaction design. User-centered design (UCD) is an approach that centers the process around the users. In this approach, design is primarily driven by the needs and objectives of users [7]. According to ISO 9241-210:2010, UCD has four main steps [8]:

1. Understanding and specifying the context of use such as user characteristics, behaviors, and objectives, related to the issues.
2. Identifying and specifying design requirements based on the context of use.
3. Implementing design solutions in the form of prototypes based on design requirements.
4. Testing and evaluating the design against the requirements.

The entire process is done iteratively, with each iteration having the potential to be repeated numerous times until the desired design is produced. To make it easier to understand, Fig. 1 explains the whole UCD process based on ISO 9241-210:2010.

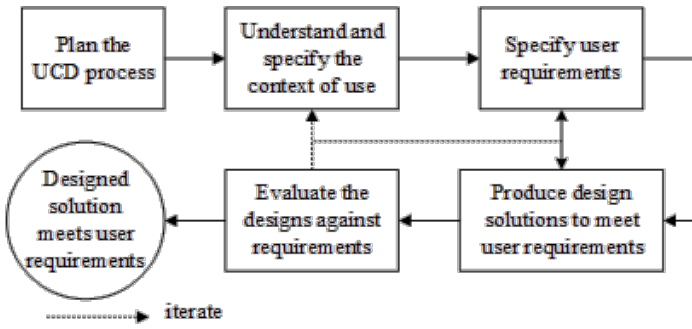


Fig. 1. UCD process.

For each iteration, the design solution is evaluated against the required usability and user experience goals. As in [6], usability refers to how an interactive product can be easy to learn, easy to use, and effective to use. There are six parameters of usability, i.e., effectiveness, efficiency, safety, utility, learnability, and memorability [6].

Meanwhile, user experience is defined as a person's views and reactions because of using or anticipating using a product or service [9]. Table I shows twelve examples of good user experience goals that can be used as a goal [6].

TABLE I. USER EXPERIENCE GOALS

Satisfying	Exciting	Challenging	Enjoyable
Entertaining	Enhancing Sociability	Engaging	Helpful
Pleasurable	Motivating	Supporting Creativity	Cognitively Stimulating

B. Usability Testing

After a design has been implemented, it should be evaluated through usability testing. Usability testing is defined as a process of evaluating the usability of a design with a group of representative users [9]. When conducting usability testing, there are several measuring methods described as follows.

1. Completion rate

The completion rate is a method to measure the effectiveness of a design by tracking how well users complete a set of tasks. The completion rate is a post-task questionnaire, which means that the completion rate is asked every time a user has done a certain task. The score given in this method is binary, with 1 indicating a user's success in completing a given task and 0 if the user fails. From a completion rate result, design effectiveness can be calculated as a ratio of the total number of completed tasks to the total number of tasks taken [10]. For a design to have good effectiveness, it should have at least a 75-90% ratio [11].

2. Single Ease Question (SEQ)

An SEQ is a method to measure the learnability of a design. Just like the completion rate, SEQ is a post-task questionnaire. After a user has done a certain task, the user is asked to rate how difficult it is to do such a task between 1 (very difficult) and 7 (very

easy). For a design to have good learnability, the minimum average SEQ value is 5.7 [12].

3. System Usability Scale (SUS)

A SUS is a method to measure the usability and learnability of a design. Unlike SEQ, SUS is a post-test questionnaire, which means that SUS is asked after a user has completed each given task. SUS contains ten questions in the form of a Likert scale with a value between 1 (strongly disagree) and 5 (strongly agree), which results in a value between 0 and 100. The formula to calculate the SUS score is:

- Odd-numbered question: answer value - 1
- Even-numbered question: 5 - answer value
- SUS score: total calculated values $\times 2.5$

For a design to have good usability and learnability, the minimum average SUS score that should be reached is 68 [13].

C. Blindness and Visual Impairment

Blindness is defined as a condition of not having vision because of injury, illness, or genetics. Meanwhile, visual impairment is defined as a specific degree of vision loss that results in problems that cannot be solved with standard solutions, such as glasses [14].

To interact with their smartphones, people with blindness and visual impairment heavily rely on assistive technology (AT). AT is described as any hardware or software that enables people to access technology. Examples of AT for blind and visually impaired people are screen readers and refreshable braille displays [15].

III. USER RESEARCH AND ANALYSIS

A. Research Method

To obtain data for this research, qualitative research is conducted in the form of interviews with blind or visually impaired students from middle to high school level in State Special Needs School A Pajajaran, Bandung. This demographic was chosen because students have the most information needs, particularly when it comes from books. The target number of participants is between 10 and 30. This range is based on finding 75% of usability problems with a minimum of 5 participants [16].

B. User Analysis

The interview managed to have 20 participants. Some points were asked during the interview such as their condition (blind or visually impaired), assistive technology they use, and their experience when using an audiobook app. The interview data will be analyzed to determine the general problems and needs of the users. From the interview, there are four problems that users have regarding audiobook applications:

1. There are very few accessible Indonesian audiobooks.
2. Some audiobook apps are either filled with ads or paywalls.
3. Some audiobook apps lack features to help visually impaired users.

- Some audiobook apps lack the basic features to use audiobooks.

To address these four problems, there are four users' needs that can be defined:

- Using Indonesian audiobooks with ease.
- Access every in-app feature without disturbance and restrictions.
- Navigate the app using assistive technology with ease.
- Search, use, and save audiobooks with full basic features.

Users' needs can be derived further to define what are users' goals and what kind of tasks users should do to achieve those goals. These user goals and tasks are listed in Table II.

TABLE II. GOALS AND TASKS OF THE USERS

Goals	Tasks
Using an audiobook	Play or resume an audiobook
	Pause an audiobook
	Increase or decrease playback speed
	Change playback volume
	Fast forward or fast rewind
Searching an audiobook	Search for the desired audiobook
Saving an audiobook	Download the desired audiobook
	Save the desired audiobook into a playlist
Using the app	Using the app with assistive technology
	Using the app without encountering any ads or paywalls
	Accessing every feature there is

User goals and tasks then will be used as a basis for determining features that should exist in the application.

C. System Analysis

The platform chosen for the design solution is the mobile platform. The reasons are:

- A mobile app can be used whenever and wherever the user wants. Therefore, users have a wide opportunity to access audiobooks.
- Most Indonesian citizens have access to mobile devices [17].

System requirements will be derived from the analyzed user analysis in the form of features that a design must have. For this application, features can be categorized into two groups, i.e., functional and non-functional.

Functional features refer to features that can be used by users to carry out the task they want to do. Functional features for this application are given in Table III.

TABLE III. FUNCTIONAL FEATURES

Goals	Features
Use an audiobook	Play an audiobook
	Pause and resume an audiobook playback
	Increase or decrease playback speed
	Change playback volume
	Fast forward or fast rewind during playback
Searching an audiobook	Search for an audiobook by typing
	Search for an audiobook by speaking
Saving an audiobook	Download an audiobook file
	Save an audiobook into a history playlist
	Save an audiobook into a favorite playlist

Meanwhile, non-functional features refer to features that can help users to access or use the functional features listed in Table III. Non-functional features for this application are:

- Dark mode.
- A descriptive label for each element
- Ad-free.
- Access to every feature.

D. Usability and User Experience Goals

Every iteration in UCD is tested against the required usability and user experience goals. Both usability and user experience goals are determined from research results, particularly user needs. Those goals are:

- Effective to use, meaning that the app must make users able to do what they are supposed to do, which is use an audiobook.
- Easy to use, meaning that the app must be easy to use by the users, particularly blind and visually impaired users.
- Helpful, meaning that users must feel helped by using this app to search, use, and save audiobooks.

IV. DESIGN IMPLEMENTATION

During the process of building the design solution, there were two iterations. Each iteration consists of creating a prototype along with usability testing. In the last iteration, the final prototype was completed. The prototype can be grouped into two main parts, i.e., the app interface and labels.

A. App Interface

The design aspects for this application are as follows:

- Color scheme
The color scheme for this app is a variation of the dark mode color scheme which consists of a dark background and bright foreground. Colors used for both background and foreground are shown in Table IV.

TABLE IV. COLOR SCHEME

Group	Color code	Example
Background	#434343	
	#121212	
Foreground	#FFFFFF	
	#B6B6B6	
	#64FFDA	

Background colors are used for container elements such as scaffolds, containers, or cards. Meanwhile, foreground colors are used for contextual elements such as texts, icons, and sliders.

Font

The font used in this app is IBM Plex Sans. This font is chosen because of its readability, e.g., the letter “I” (capital letter “i”) and “l” (small letter “L”) are easily distinguishable.

In this application, there are five views, i.e., main, search, list, book, and play views.

1. Main View

Main View consists of three tabs, i.e., home, history, and favorites tabs. The main purpose of the Main View is to provide quick access to audiobooks. This is evident in a variety of audiobook cards in either the home, history, or favorites tab. By tapping on this card, users will be redirected to the book view. From the Main View, users can also go to search and list view by tapping the search icon and three horizontal dots icon, respectively. The Main View is displayed in Fig. 2.

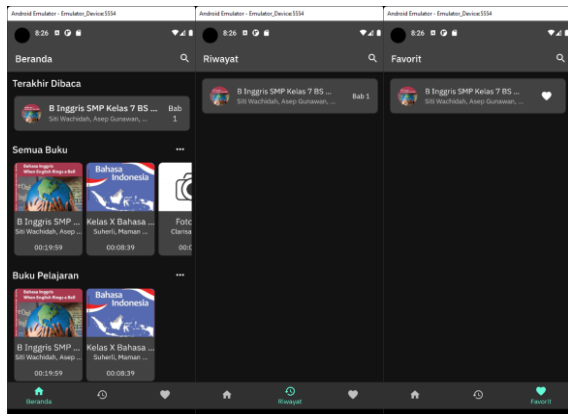


Fig. 2. Main View

2. Search View

The main purpose of Search View is to allow users to look for specific audiobooks. Users can search for audiobooks by giving out keywords. Keywords can be entered either by typing, tapping the system keyboard, or by speaking, which is done by tapping the mic icon and speaking the keywords verbally. Search View is displayed in Fig. 3.

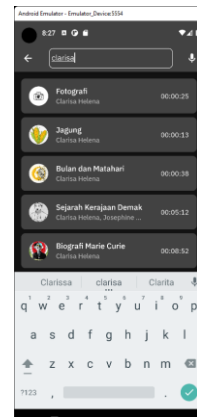


Fig. 3. Search View

3. List View

The main purpose of List View is to access audiobooks based on specific categories. The audiobooks listed in this view can be changed, depending on the category selected in the main view. List View is displayed in Fig. 4.

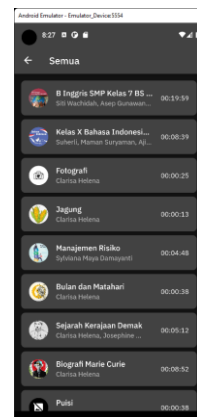


Fig. 4. List View

4. Book View

The main purpose of Book View is to display information about an audiobook to the users. Information includes titles, authors, genres, descriptions, and a list of chapters. Users can also select which chapter to play from this view by tapping the desired chapter card from the list of chapters. This action will also redirect users to Play View. Book View is displayed in Fig. 5.

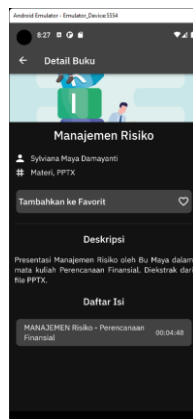


Fig. 5. Book View

5. Play View

The main purpose of Play View is to allow users to control the current playback. From this view, users can pause, resume, fast forward, fast rewind, go to the previous or next chapter, and change playback speed or volume by using icon buttons or sliders. Users can also download the audio file and save the current book to favorites. Play View is displayed in Fig. 6.

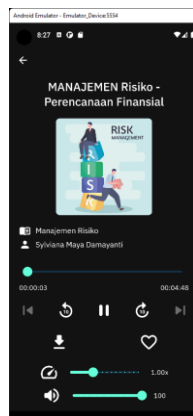


Fig. 6. Play View

B. Labels

Every important element, such as buttons and cards, in the app, is labeled to help users that use screen readers easily understand the functions of each element. Both buttons and cards have their label format.

There are two types of cards, a card about books and a card about the chapter. In general, both cards contain information about the title, authors (books only), duration, and usage hints. The label format for both cards is “<title>, Penulis: <authors>, Durasi: <duration>, Ketuk dua kali untuk <usage hint>.”

Meanwhile, the label for the button contains two pieces of information: a short description of such button and a usage hint. In general, the format label for each button is “<description>, Tombol, Ketuk dua kali untuk <usage hint>.”

To make it easier to imagine, look at Fig. 7 below, which shows how the screen looks when a user is accessing the application with TalkBack, a screen reader in Android.

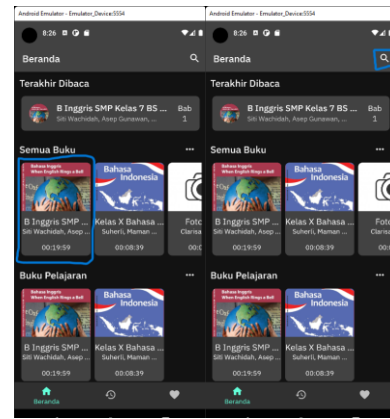


Fig. 7. Accessing App with TalkBack

The first picture in Fig. 7 shows how the screen looks when the user is focused on a book card. When this happens, the screen reader will describe the focused element to the user by using the defined label. In this case, the screen reader will say “B Inggris SMP..., Penulis: Siti Wachidah, ..., Durasi: 0 jam, 19 menit, 59 detik, Ketuk dua kali untuk membuka buku B Inggris SMP...”

The same thing is applied to the second picture, which shows how the screen looks when the user is focused on the search button. When this happens, the screen reader will say “Pencarian, Tombol, Ketuk dua kali untuk membuka halaman pencarian.”

A demonstration video can be watched from this link to learn about labels for other elements in the app¹.

V. EVALUATION

To evaluate the designed application, it is tested against 5 participants with both blindness and visual impairment. Each participant is asked to do 6 tasks. The descriptions for those tasks are:

1. Access an audiobook with a specific category.
2. Search for an audiobook by typing and speaking.
3. Access information is given about an audiobook.
4. Play an audiobook and control the current playback.
5. Save an audiobook by adding it to the list or downloading the file.
6. Quick access to the saved audiobook in the list.

To measure usability and user experience goals, several measuring methods were used. The mapping between goals and measurement method used is given in Table V.

TABLE V. MEASUREMENT METHODS

No.	Goals	Measurement Method
1	Effective to use	Completion rate
2	Easy to use	SEQ, SUS
3	Helpful	Likert-scale, questionnaire

Based on usability testing results, the results for each goal are explained as follows.

1. Effective to use (effectiveness)

¹ https://bit.ly/my_audiobook

According to the completion rate results, all 5 participants completed 100% of the tasks assigned. Because the completion rate ratio exceeds 75-90%, it can be concluded that the designed application has good effectiveness.

2. Easy to use (learnability)

Learnability is evaluated by two measures, i.e., average SEQ and average SUS. Based on SEQ results, the average SEQ score is 6.97 on a scale between 1 and 7. Meanwhile, based on SUS results, the average SUS score is 92.5 on a scale between 0 and 100.

Both average SEQ and SUS scores have exceeded minimum average scores, i.e., 5.7 and 68, respectively. Therefore, it can be concluded that the designed application has good learnability and usability.

3. Helpful

Helpfulness is evaluated with a Likert scale on a scale between 1 and 5. Based on the results, the average Likert-scale value is 5 on a scale of 5. Therefore, it can be concluded that all participants feel that the designed application is very helpful.

VI. CONCLUSION AND FUTURE WORK

The development of audiobook application interaction design for visually impaired people has been discussed based on the User-Centered Design approach. The entire process is done iteratively. For each iteration, the design solution is evaluated against the required usability and user experience goals. After a design has been implemented, it was evaluated through usability testing. Based on the results, it has been concluded that the designed application has met all usability and user experience goals. Therefore, it can be inferred that the designed application is appropriate to be used by blind and visually impaired users to access audiobooks. Good color contrast, readable font, descriptive labels to explain functions for elements on the screen, and less intrusive element such as ads in the application are some properties of the designed application that make it accessible to blind and visually impaired users. For future work, it can be considered to change the words *Last Read* to *Last Listened* for a better context.

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